

DG INTERNS HUB

Cybersecurity Internship Program

WEEK 3

ADVANCED NETWORK SECURITY ASSESSMENT & TRAFFIC INVESTIGATION

Prepared By: Prakhar gupta
Role: Cybersecurity Intern
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3. Executive Summary

This report documents an authorized network security assessment and traffic investigation performed against a single Windows-based target host (192.168.1.10) within an isolated VirtualBox lab environment, as part of Week 3 of the Cybersecurity Internship programme. The purpose of the assessment was to discover live hosts and exposed services,

analyse those services for version and configuration weaknesses, capture and interpret live network traffic with Wireshark, correlate Nmap scan activity with the corresponding packets on the wire, identify security findings supported by evidence, and apply firewall-based hardening to reduce the target's attack surface.

Scanning from an external (Kali) vantage point using stealth/connect scans initially showed only one TCP port — 4899 (Radmin remote administration) — as reachable, with the remaining 999 ports reported as filtered. However, scans executed from a local/administrative session on the target itself revealed a substantially larger set of listening services: 135 (MSRPC), 139 (NetBIOS-SSN), 445 (Microsoft-DS/SMB), 4899 (Radmin), 5432 (PostgreSQL), 8000 (HTTP — Splunk daemon), and 8089 (SSL/HTTP — Splunk daemon). This discrepancy is itself a key finding: a perimeter/host firewall is filtering most inbound ports from the network, while the services remain fully active and reachable to any process or scan executed locally on the host.

Wireshark captures over the course of the assessment recorded background DNS resolution, ARP address resolution, ICMP connectivity checks to the default gateway, and substantial outbound TLS/QUIC traffic to cloud and telemetry endpoints (Google, Microsoft, Splunk-related, and other services). Filtering on TCP control flags allowed the SYN, SYN/ACK, and RST/ACK phases of numerous TCP handshakes to be isolated and correlated directly with the Nmap scans that were run in parallel, confirming that Nmap's reported states are consistent with the packets actually observed on the wire.

An NSE vulnerability scan (`--script vuln`) identified the Splunk HTTP service on port 8000 as "LIKELY VULNERABLE" to the Slowloris denial-of-service condition (CVE-2007-6750), and enumerated a reachable `/robots.txt` on the same service. SMB-focused vulnerability checks (`ms10-061`, `cve-2012-1182`, `ms10-054`) did not return conclusive results due to protocol negotiation failures during the scan, and are recorded as inconclusive rather than confirmed-safe or confirmed-vulnerable.

Hardening was applied on the target using the native Windows Firewall (`netsh advfirewall`) by creating explicit inbound block rules for the three most exposed and least-necessary services — Radmin (4899/TCP), PostgreSQL (5432/TCP), and the Splunk HTTP interface (8000/TCP). The rules were verified as present and enabled with an action of "Block" for all profiles (Domain, Private, Public). A verification scan run immediately afterwards — again from the local/administrative session on the target itself — continued to report these ports as open. This is recorded transparently in this report as an evidence-based observation rather than a definitive hardening failure: scans launched from the same host they target commonly traverse the local network stack in a way that does not pass through the host's own inbound firewall filtering path, so the local scan is not, by itself, sufficient evidence of whether the block rules are effective against genuinely external traffic. This distinction is discussed further in Sections 20, 23, and 26.

Overall, the assessment confirms a working, evidence-driven methodology: authorized discovery, layered Nmap analysis, packet-level verification with Wireshark, a documented vulnerability finding, and a concrete (if only partially verifiable from the available vantage point) hardening action. The recommendations in Section 18 focus on re-testing the applied firewall rules from an external vantage point, reducing the number of unnecessarily internet/networkfacing administrative services, and patching or reconfiguring the Splunk HTTP service against the identified Slowloris exposure.

4. Introduction

A network security assessment is the structured process of identifying the hosts, services, and traffic patterns present on a network and evaluating them against known security weaknesses and best practice. It is a foundational activity in both offensive (penetration testing) and defensive (blue team / hardening) security work, because an organisation cannot secure what it has not first discovered and understood.

Network discovery matters because unmanaged or forgotten hosts and services are one of the most common sources of real-world compromise: a service that administrators are unaware is running cannot be patched, monitored, or restricted. Identifying exposed ports and the services bound to them allows an analyst to reason about whether that exposure is necessary for the host's function, and if so, whether it is adequately protected.

Packet-level analysis with a tool such as Wireshark complements port/service scanning by showing what is actually happening on the wire — confirming or refuting assumptions drawn purely from scan output, revealing background traffic (DNS, ARP, ICMP) that scanning alone would not show, and allowing scan activity itself to be correlated against the exact TCP flags exchanged during connection attempts.

Finally, reducing the attack surface — by disabling unnecessary services, closing or firewalling unneeded ports, and applying vendor patches — is the practical objective that discovery and analysis exist to support. This report follows that full chain: discovery, analysis, traffic capture, correlation, findings, and hardening.

5. Objectives

- Discover authorised hosts within the lab network and confirm the target is reachable.
- Identify exposed ports and services running on the target host using Nmap.
- Perform layered Nmap analysis: service/version detection, OS fingerprinting, TCP SYN scanning, UDP scanning, and NSE vulnerability scripts.
- Capture live network traffic using Wireshark across multiple protocols (DNS, ARP, ICMP, TCP/TLS/QUIC).
- Analyse captured packets to understand communication patterns and protocol behaviour.
- Compare and correlate Nmap scan results with the corresponding Wireshark packet evidence.
- Investigate specific network events of interest, including TCP handshakes, resets, and unusual local connection attempts.
- Identify and document security findings supported directly by the collected evidence.
- Apply firewall-based hardening to reduce the exposed attack surface of the target.
- Verify the effect of the hardening changes and honestly record the limitations of that verification.

6. Lab Environment

The lab environment consisted of an analyst machine running Kali Linux (used for the initial external Nmap scans and all Wireshark packet captures) and a Windows-based target virtual machine at 192.168.1.10, both connected to the same

local network segment (192.168.1.0/24, gateway 192.168.1.1). Later scans and the hardening activity were performed directly from an administrative command-line session on the target itself.

Component	Details
Analyst Operating System	Kali Linux 2025.3 (VirtualBox guest)
Analyst Tools	Nmap 7.95 (Kali), Nmap 7.98 (Windows target), Wireshark
Target Host	192.168.1.10 — Windows host (MAC 9C:C7:D3:2D:94:B8)
Default Gateway	192.168.1.1
Network Type	Local IPv4/IPv6 dual-stack network, 192.168.1.0/24
Virtualization Platform	Oracle VirtualBox (VM: kali-linux-2025.3-virtualbox-amd64)
Authorization Scope	Isolated internal lab network; target host and IP range explicitly assigned for this internship exercise

7. Network Topology

No dedicated network topology diagram was captured as part of the submitted evidence. Based on the addressing and MAC information observed consistently across the Nmap and Wireshark evidence, the lab network can be described as follows: a single local segment (192.168.1.0/24) with a default gateway at 192.168.1.1; the analyst/Kali machine and the Windows target (192.168.1.10, MAC 9C:C7:D3:2D:94:B8) reside on this same segment; ARP traffic captured in Wireshark additionally shows several other hosts active on the same broadcast domain (for example 192.168.1.74, 192.168.1.100, 192.168.1.57, and others), consistent with a shared Wi-Fi/LAN segment rather than a dedicated point-to-point link. [Evidence Not Provided: a diagram or screenshot of physical/virtual network adapter topology.]

8. Tools Used

Tool	Purpose
Nmap (Kali, v7.95)	Host discovery, port scanning, service/version detection, OS fingerprinting, NSE vulnerability scanning — run from the external analyst machine
Nmap (Windows target, v7.98)	Local verification scans and vulnerability scripting run directly on the target host
Wireshark	Live packet capture and protocol analysis (DNS, ARP, ICMP, TCP, TLS, QUIC) on the Wi-Fi interface

Windows Command Prompt / netsh advfirewall

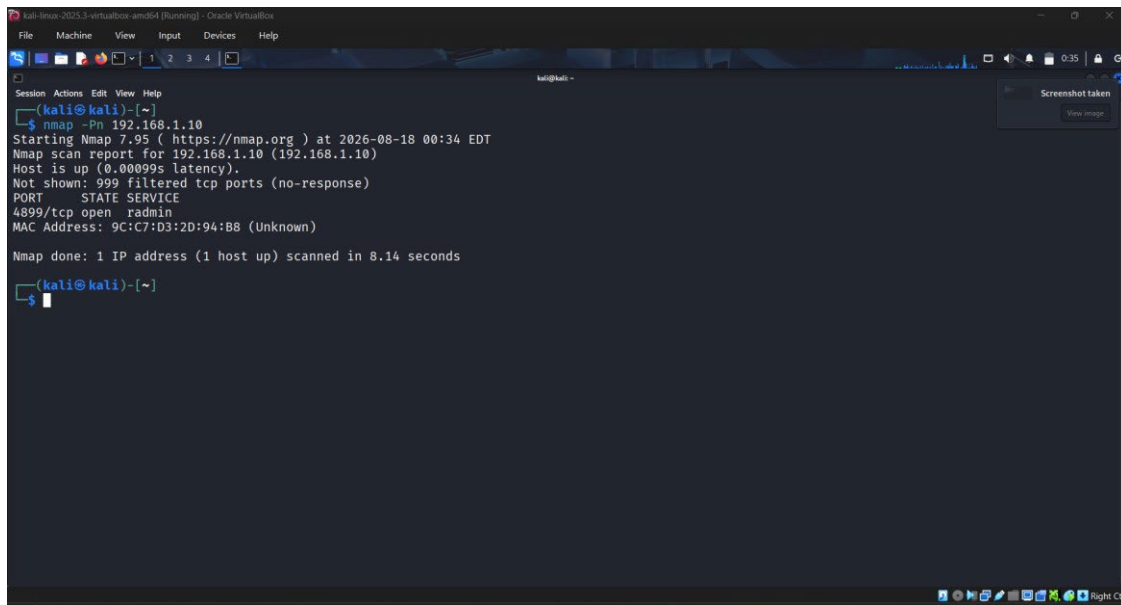
Creation and verification of Windows Firewall inbound block rules for hardening

9. Task 7 – Network Discovery & Basic Nmap Scanning

The objective of this task was to confirm the target host was reachable and to run an initial, low-noise Nmap scan to identify any immediately visible open ports before deeper analysis.

9.1 Basic Host Discovery Scan (nmap -Pn)

A ping-suppressed scan (-Pn) was run against 192.168.1.10 to force Nmap to scan all ports regardless of ICMP reachability. The scan completed in 8.14 seconds and reported the host up with 999 ports filtered (no response) and a single open TCP port: 4899 (radmin). The target's MAC address (9C:C7:D3:2D:94:B8) was also captured.



```
kali@kali:~$ nmap -Pn 192.168.1.10
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 00:34 EDT
Nmap scan report for 192.168.1.10 (192.168.1.10)
Host is up (0.00099s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
4899/tcp  open  radmin
MAC Address: 9C:C7:D3:2D:94:B8 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 8.14 seconds

kali@kali:~$
```

Figure NMAP 1.1 – Basic Nmap Host Discovery Scan (nmap -Pn 192.168.1.10)

9.2 Service/Version Confirmation (nmap -sV -Pn)

A follow-up service/version detection scan confirmed that port 4899/tcp is running Famatech Radmin 3.0, and Nmap's service-info fingerprinting identified the underlying operating system family as Windows (CPE cpe:/o:microsoft:windows). This scan took 13.95 seconds.

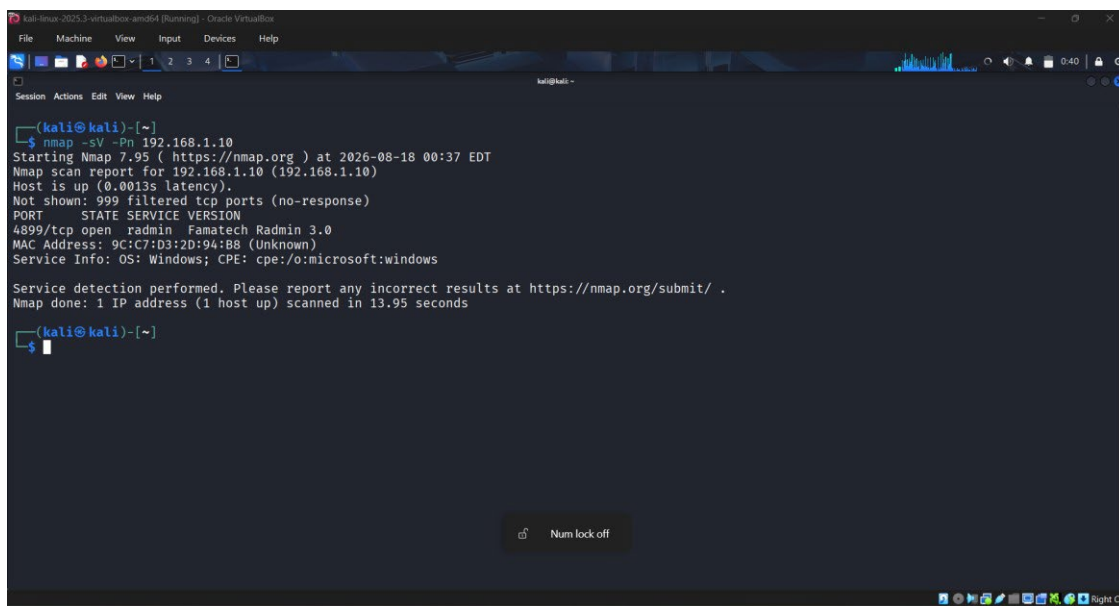


Figure NMAP 1.2 – Service and Version Detection Confirming Radmin 3.0 on Port 4899

9.3 Background DNS Activity Observed During Discovery

In parallel with the discovery scans, a Wireshark capture filtered on "dns" was reviewed to understand ordinary background traffic on the network segment. The capture (20,154 total packets, 240 matching the DNS filter) showed routine DNS resolution for common cloud/service domains (ssl.gstatic.com, play.google.com, clientservices.googleapis.com), Nessus-related lookups (plugins.nessus.org), and a reverse DNS (PTR) query resolving 10.1.168.192.in-addr.arpa to 192.168.1.10, confirming the target's own reverse-DNS identity on the segment.

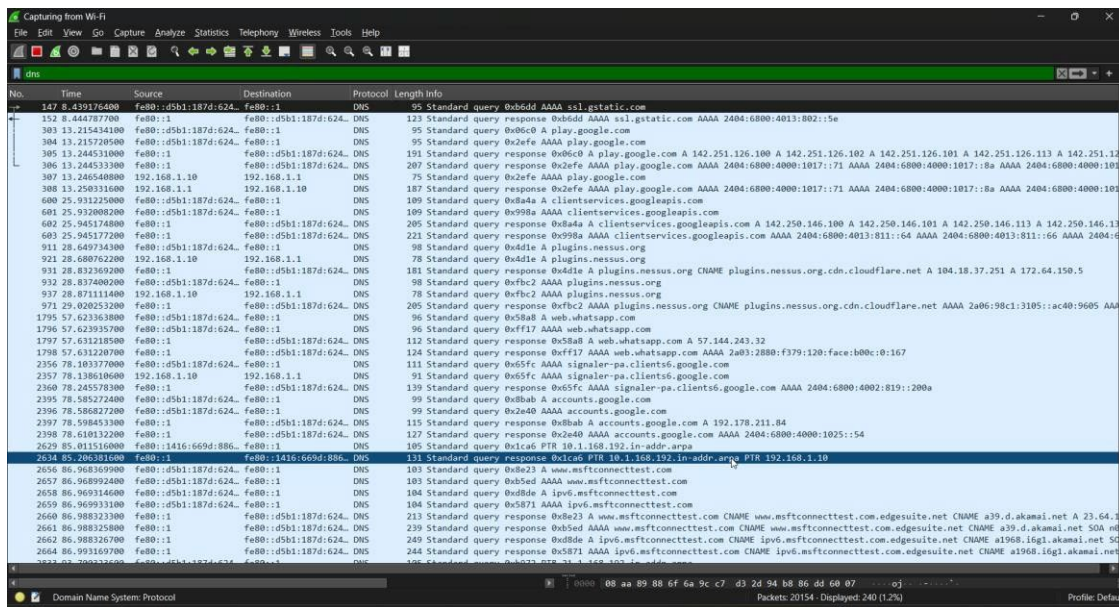


Figure WIRESHARK 1.1 – DNS Query Activity Captured During the Discovery Phase

9.4 Task 7 Summary Table

IP Address	Status	Open Ports	Protocol	Service	Observation
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192.168.1.10	Up	4899	TCP	radmin (Famatech Radmin 3.0)	Only port visible from external scan vantage point; 999 ports filtered
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10. Task 8 – Advanced Nmap Security Assessment

Building on the basic discovery scan, a series of deeper Nmap techniques were run against the same target to obtain OS fingerprinting, confirm reachability using an alternate scan type, enumerate UDP services, and check for known vulnerabilities via the Nmap Scripting Engine (NSE).

10.1 Repeat Service/Version Detection

The service/version scan was repeated for verification and produced the same result as Task 7: port 4899/tcp open, running Famatech Radmin 3.0, on a Windows host.

```
kali@kali:~$ nmap -sV -p4899 192.168.1.10
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 00:37 EDT
Nmap scan report for 192.168.1.10 (192.168.1.10)
Host is up (0.0013s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
4899/tcp  open  radmin  Famatech Radmin 3.0
MAC Address: 9C:C7:D3:2D:94:B8 (Unknown)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.95 seconds

kali@kali:~$
```

Figure NMAP 2.1 – Repeated Service/Version Detection (Verification)

10.2 Operating System Detection (nmap -O)

An OS detection scan was attempted. Nmap issued a warning that OS results may be unreliable because it could not find at least one open and one closed port to fingerprint against (a direct consequence of the surrounding ports being reported as filtered rather than closed). Nmap's best guesses ranged across Windows 11 21H2 (91%), FreeBSD 6.2RELEASE (88%), Windows Server 2008 (85%), and Windows 10 1607 (85%), with no exact match found. This result should be treated as low-confidence and is reported as such rather than as a confirmed OS.

```

kali@kali:~$ nmap -sS 192.168.1.10
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 03:02 EDT
Nmap scan report for 192.168.1.10 (192.168.1.10)
Host is up (0.0012s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
4899/tcp  open  radmin
OS: Linux 3.10 (Ubuntu 12.04 LTS)
OS CPE: cpe:/o:linux:3.10
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 25.50 seconds

```

Figure NMAP 2.2 – OS Detection (Low-Confidence Guesses Due to Filtered Ports)

10.3 TCP SYN (Stealth) Scan (nmap -sS)

A SYN scan was run as an alternative, lower-noise scan technique. It returned the same result as the connect-based scans — port 4899/tcp open (radmin), 999 filtered — completing in 17.25 seconds. This consistency across scan techniques increases confidence that 4899/tcp is genuinely open and reachable from the external vantage point, rather than being an artefact of one particular scan method.

```

kali@kali:~$ nmap -sS 192.168.1.10
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 03:13 EDT
Nmap scan report for 192.168.1.10 (192.168.1.10)
Host is up (0.00004s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
4899/tcp  open  radmin
OS: Linux 3.10 (Ubuntu 12.04 LTS)
OS CPE: cpe:/o:linux:3.10
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.25 seconds

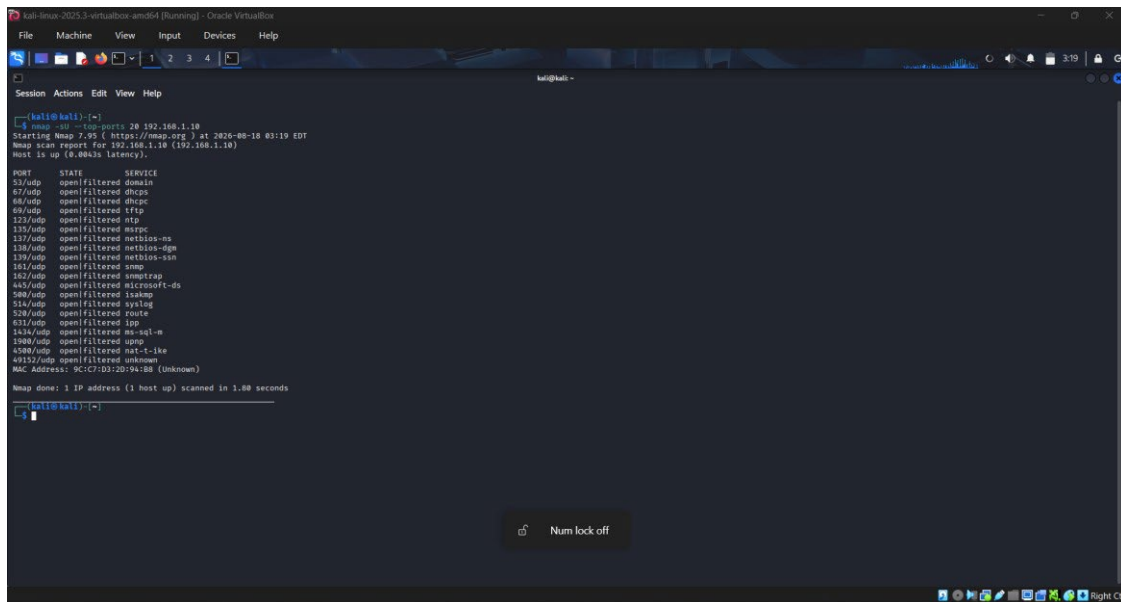
```

Figure NMAP 2.3 – TCP SYN Stealth Scan (nmap -sS)

10.4 UDP Scan – Top 20 Ports (nmap -sU --top-ports 20)

A UDP scan of the 20 most common UDP ports was performed. All 20 ports (53/domain, 67/dhcp, 68/dhpcp, 69/tftp, 123/ntp, 135/msrpc, 137/netbios-ns, 138/netbios-dgm, 139/netbios-ssn, 161/snmp, 162/snmptrap, 445/microsoft-ds, 500/isakmp, 514/syslog, 520/route, 631/ipp, 1434/ms-sql-m, 1900/upnp, 4500/nat-t-ike, 49152/unknown) were reported as open|filtered. This is Nmap's standard response for UDP ports that neither returned a service response nor

an ICMP port-unreachable message, and does not, by itself, confirm that all of these UDP services are actually running — it primarily reflects that ICMP unreachable responses are being suppressed or rate-limited on the path to the host.



```
kali@kali:~$ nmap -sU --top-ports 20 192.168.1.10
Starting Nmap 7.95 ( https://nmap.org ) at 2018-08-18 03:19 EDT
Nmap scan report for 192.168.1.10 (192.168.1.10)
Host is up (0.0042s latency).

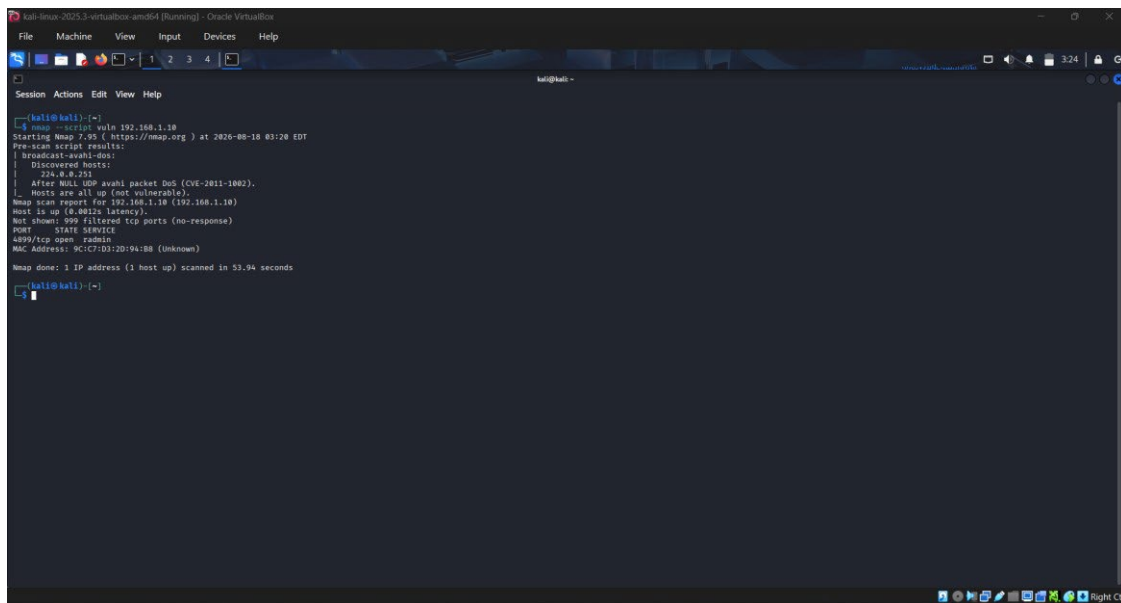
PORT      STATE      SERVICE
22/udp    open|filtered domain
67/udp    open|filtered dhcp
68/udp    open|filtered dhcp
69/udp    open|filtered rtp
123/udp   open|filtered ntp
135/udp   open|filtered mapi
137/udp   open|filtered netbios-ns
138/udp   open|filtered netbios-dgm
139/udp   open|filtered netbios-ssn
161/udp   open|filtered snmp
162/udp   open|filtered snmptrap
445/udp   open|filtered microsoft-ds
280/udp   open|filtered rsync
314/udp   open|filtered syslog
520/udp   open|filtered route
611/udp   open|filtered lrp
1434/udp  open|filtered ms-sql-m
1500/udp  open|filtered udp
4899/udp  open|filtered nat-t-like
8015/tcp  open|filtered unknown
NMC Address: MC(C)10120194188 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 1.88 seconds
```

Figure NMAP 2.4 – UDP Scan of the Top 20 Ports (*nmap -sU --top-ports 20*)

10.5 NSE Vulnerability Pre-Scan (*nmap --script vuln*)

An NSE vulnerability scan was run. Its pre-scan broadcast-avahi-dos check sent a NULL UDP Avahi packet (associated with CVE-2011-1002) to the local multicast address 224.0.0.251 and confirmed that hosts on the segment, including the target, remained up and were not vulnerable to that specific denial-of-service condition. The main scan against 192.168.1.10 again returned only port 4899/tcp open with 999 filtered, so the broader NSE vulnerability script set had only that single filtered/open surface to test against from this vantage point, and the scan completed in 53.94 seconds without additional findings at this stage (a more targeted, port-specific vulnerability scan was later performed in Task 12 once the additional locally-visible ports had been identified).



```
kali@kali:~$ nmap --script vuln 192.168.1.10
Starting Nmap 7.95 ( https://nmap.org ) at 2018-08-18 03:28 EDT
Pre-scan script results:
| broadcast-avahi-dos:
|   Discovered hosts:
|     224.0.0.251
|   After NULL UDP avahi packet Dos (CVE-2011-1002).
|   Hosts are all up (not vulnerable).
|_ Nmap scan report for 192.168.1.10 (192.168.1.10)
Host is up (0.0012s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE      SERVICE
4899/tcp  open      radmin
NMC Address: MC(C)10120194188 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 53.94 seconds
```

Figure NMAP 2.5 – NSE Vulnerability Pre-Scan (*nmap --script vuln*)

10.6 Task 8 Summary Table

IP Address	Port	Protocol	Service	Version	Risk	Observation
192.168.1.10	4899	TCP	radmin	Famatech Radmin 3.0	Medium	Remote administration service reachable from external scan; consistent across scan types
192.168.1.10	20 UDP ports	UDP	Various (DNS, DHCP, NetBIOS, SNMP, etc.)	[Evidence Not Provided]	Low–Medium	Reported open filtered; not individually confirmed as active services

10.7 Service Analysis: Radmin (4899/TCP)

- What service is running? Famatech Radmin 3.0, a proprietary remote-desktop/remote-administration service for Windows.
- Why may it be required? Remote administration tools are commonly used by IT staff to manage machines without physical access.
- Is exposure necessary? Exposure to the general network segment (rather than a restricted management VLAN or VPN) is not necessary for normal end-user operation of the host.
- Potential security impact: Radmin has a documented history of authentication-related and brute-force weaknesses in older releases; unrestricted network exposure of a remote-administration listener meaningfully increases the attack surface for credential attacks and unauthorized remote access.
- Recommended security control: Restrict access to 4899/TCP to a management subnet or VPN, enforce strong authentication, and keep Radmin patched to its latest release; alternatively disable the service if remote administration is not actively required (see Task 13 hardening).

11. Task 9 – Wireshark Network Traffic Capture

This task captured live traffic on the Kali analyst machine's Wi-Fi interface to observe the protocols actually present on the network segment, independent of the Nmap scan results.

11.1 DNS Traffic Capture

A DNS-filtered capture (20,154 total packets; 240 displayed) showed a mix of standard queries and responses for Google infrastructure (play.google.com, clientservices.googleapis.com), security-tooling domains (plugins.nessus.org, resolving via CNAME to plugins.nessus.org.cdn.cloudflare.net), messaging services (web.whatsapp.com), and Microsoft connectivity checks (www.msftconnecttest.com, ipv6.msftconnecttest.com). A PTR lookup for 10.1.168.192.inaddr.arpa resolved to 192.168.1.10, confirming the target's reverse-DNS record.

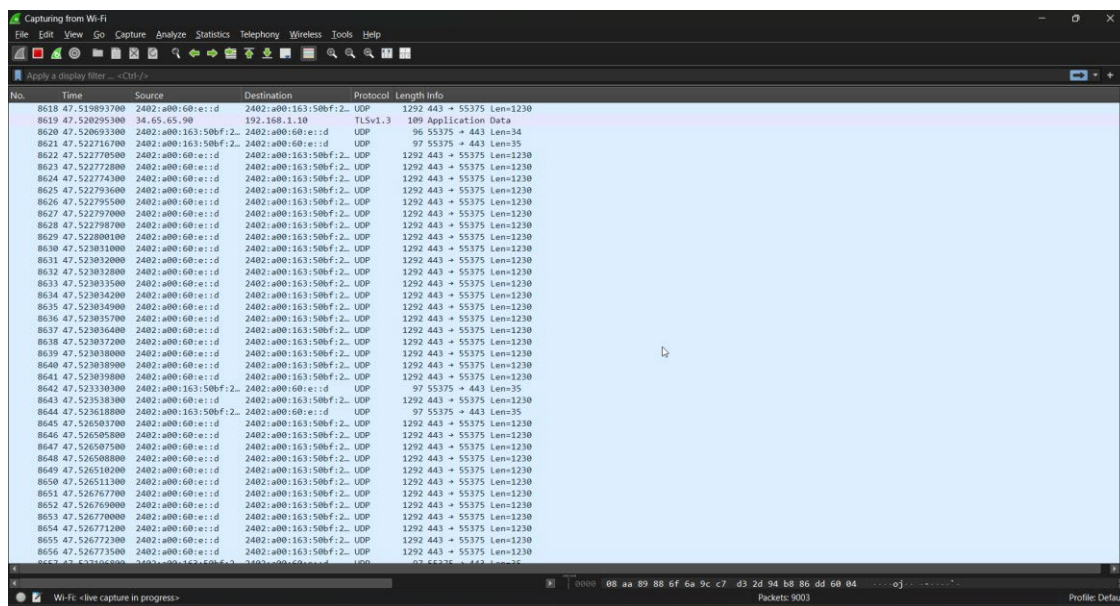


Figure WIRESHARK 2.1 – DNS Query and Response Capture

11.2 Live Capture – TLS/QUIC Application Traffic

An unfiltered live capture (9,003 packets) showed sustained UDP traffic on port 443 between the target's IPv6 address and a Google/cloud IPv6 range, consistent with QUIC (HTTP/3), alongside a TLSv1.3 Application Data exchange with IPv4 host 34.65.65.90. This illustrates that a large share of ordinary background traffic on the segment is encrypted application data rather than plaintext protocols.

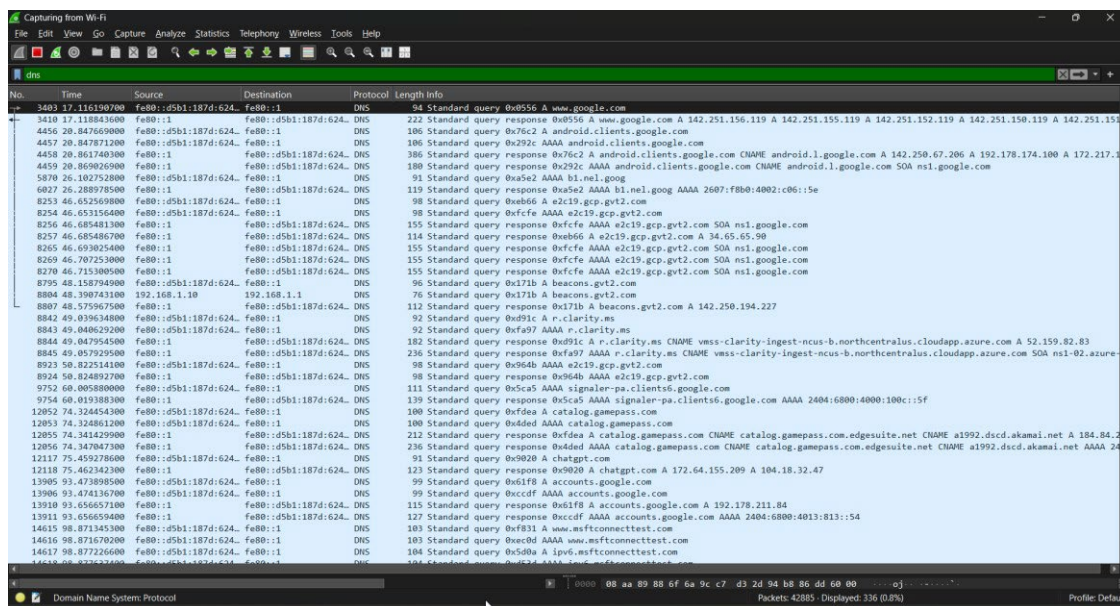


Figure WIRESHARK 2.2 – Unfiltered Live Capture Showing TLS/QUIC Application Traffic

11.3 Extended DNS Capture

A second, longer DNS capture (42,885 total packets; 336 displayed) captured additional resolution activity, including android.clients.google.com, e2c19.gcp.gvt2.com, beacons.gvt2.com, r.clarity.ms (resolving via CNAME into vmssclarity-ingest-ncus-b.northcentralus.cloudapp.azure.com — Microsoft Clarity analytics), catalog.gamepass.com, chatgpt.com, and accounts.google.com. This confirms a broad mix of ordinary consumer/cloud application traffic on the monitored segment.

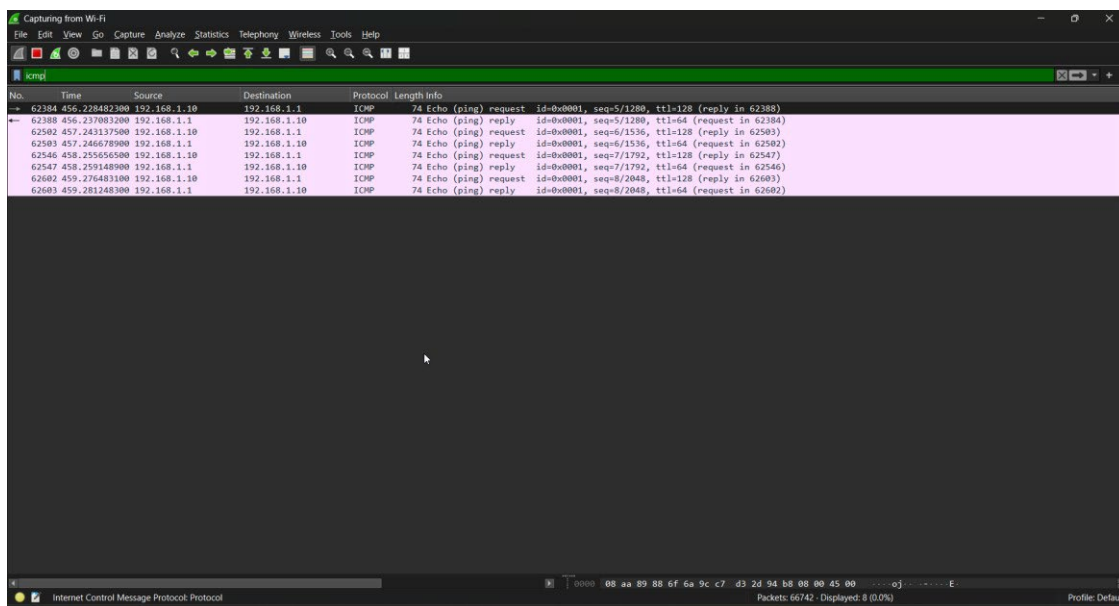


Figure WIRESHARK 2.3 – Extended DNS Capture With Additional Domains Resolved

11.4 ICMP Traffic – Gateway Reachability

An ICMP-filtered view (66,742 total packets; 8 displayed) isolated echo request/reply pairs between the target (192.168.1.10) and the default gateway (192.168.1.1), with request TTL=128 and reply TTL=64, confirming routine connectivity/liveness checking between the host and its gateway.

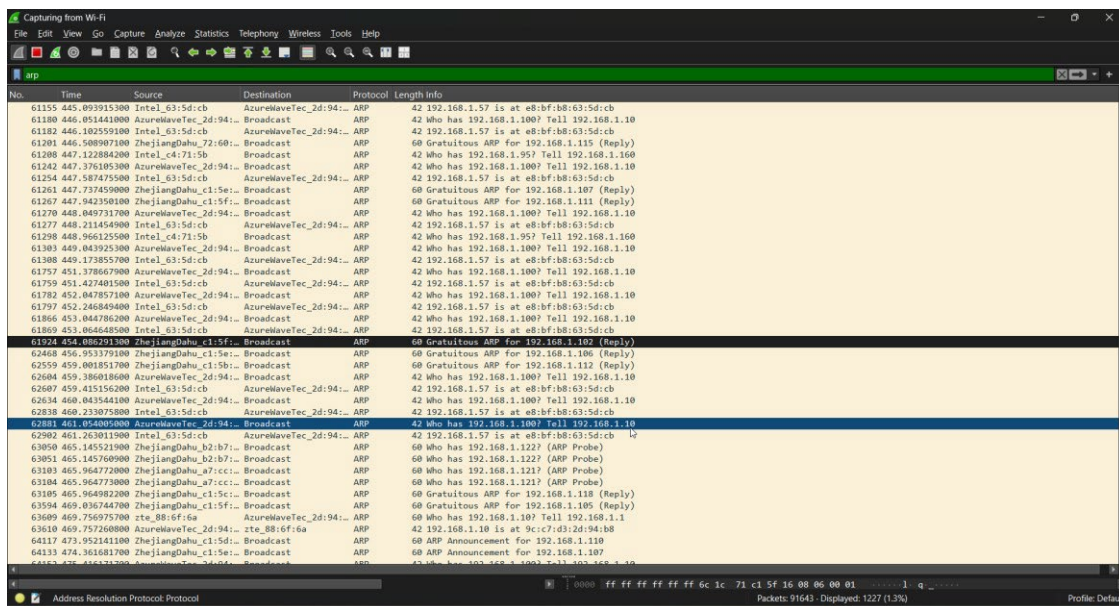


Figure WIRESHARK 2.4 – ICMP Echo Request/Reply Traffic to the Default Gateway

11.5 ARP Traffic – Local Address Resolution

An ARP-filtered view (91,643 total packets; 1,227 displayed) captured extensive local address-resolution activity: repeated "Who has 192.168.1.100? Tell 192.168.1.10" broadcasts from the target, gratuitous ARP announcements from several other hosts on the segment, and — notably — a broadcast "Who has 192.168.1.10? Tell 192.168.1.1" answered directly by the target with "192.168.1.10 is at 9c:c7:d3:2d:94:b8", which independently confirms the target's MAC address seen throughout the Nmap evidence.

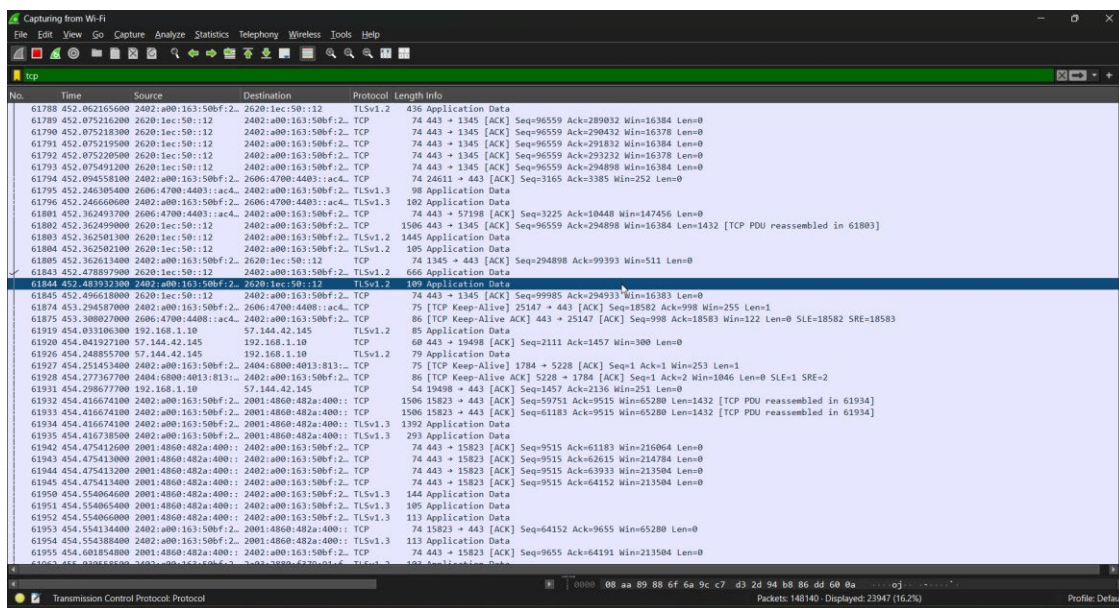


Figure WIRESHARK 2.5 – ARP Traffic Confirming the Target's MAC Address

12. Task 10 – Nmap + Wireshark Investigation

This task correlated the Nmap scanning activity with the corresponding packets captured on the wire, and separately captured a locally-run Nmap scan on the target itself to compare against the externally-observed results.

12.1 TCP Application Data Capture

A TCP-filtered capture (148,140 total packets) recorded a large number of established TLS 1.2/1.3 sessions between the target and a wide range of IPv6 destinations, each showing the expected steady-state ACK/Application-Data pattern of an already-established connection, rather than scan traffic.

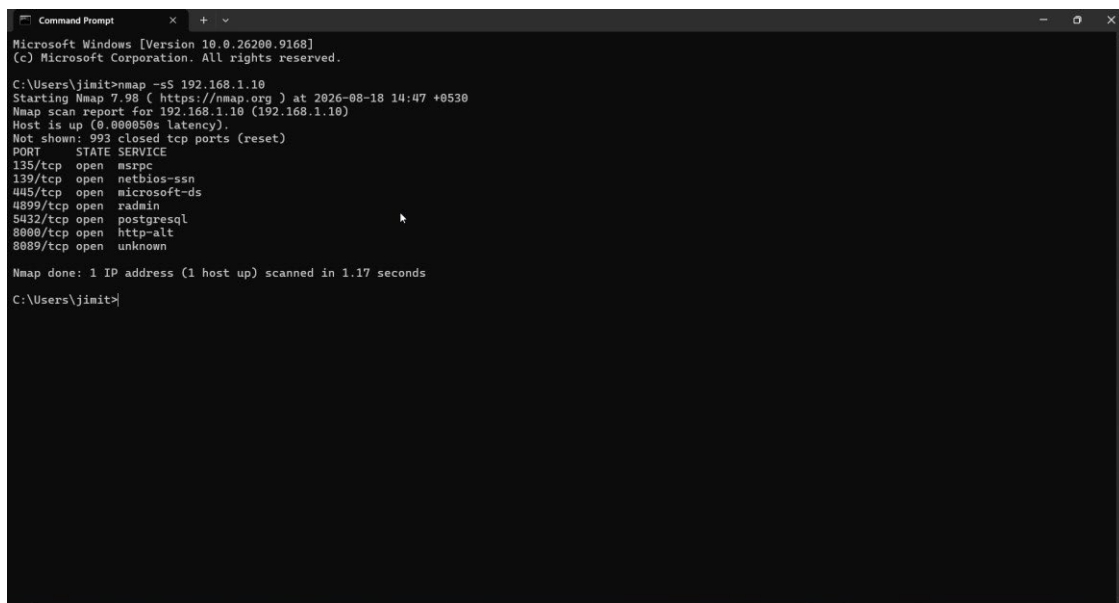


Figure NMAP-WIRESHARK 1.1 – TCP Application Data Sessions Captured on the Wire

12.2 Locally-Run SYN Scan on the Target

A SYN scan (nmap -ss 192.168.1.10) was additionally executed directly from a Windows Command Prompt on the target host itself. Run this way, the scan reported 993 closed ports (reset) and seven open ports: 135/msrpc, 139/netbios-ssn, 445/microsoft-ds, 4899/radmin, 5432/postgresql, 8000/http-alt, and 8089/unknown — completing in 1.17 seconds. This is a materially different (and more complete) picture than the same target scanned externally from Kali, where only 4899/tcp appeared open and the remaining 999 ports were filtered rather than closed. The most likely explanation is that a perimeter or host-based firewall is filtering inbound traffic to most of these ports from the network, while the services themselves are genuinely bound and listening on the host — a distinction that is only visible by comparing the external and local scan results side by side.

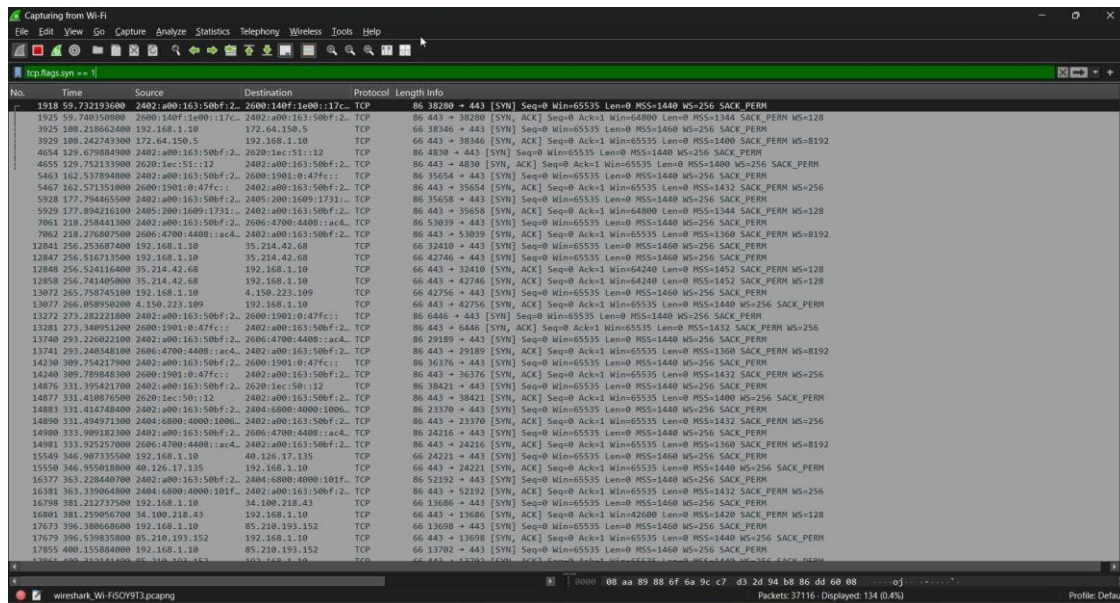


Figure NMAP-WIRESHARK 1.2 – SYN Scan Executed Locally on the Target, Revealing Additional Listening Services

12.3 SYN Packets Captured (tcp.flags.syn==1)

Filtering the capture on tcp.flags.syn==1 (37,116 total packets; 134 displayed) isolated the connection-initiation half of numerous TCP handshakes, predominantly outbound HTTPS connections (destination port 443) from the target to a wide variety of external IPv4/IPv6 addresses, each showing standard SYN options (MSS, window scaling, SACK permitted).

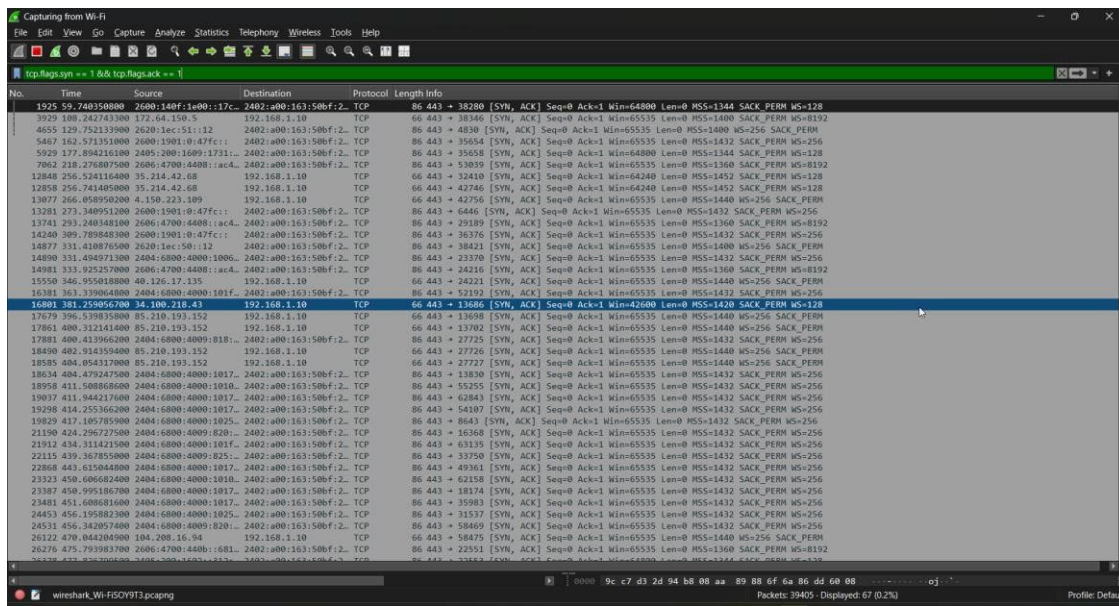


Figure NMAP-WIRESHARK 1.3 – SYN Packets Isolated via Display Filter

12.4 SYN/ACK Packets Captured (tcp.flags.syn==1 && tcp.flags.ack==1)

Refining the filter to `tcp.flags.syn==1 && tcp.flags.ack==1` (39,405 total packets; 67 displayed) isolated only the server's half of the handshake, confirming that the corresponding SYN packets observed above were in fact answered and that full three-way handshakes were completing successfully for these outbound HTTPS sessions.

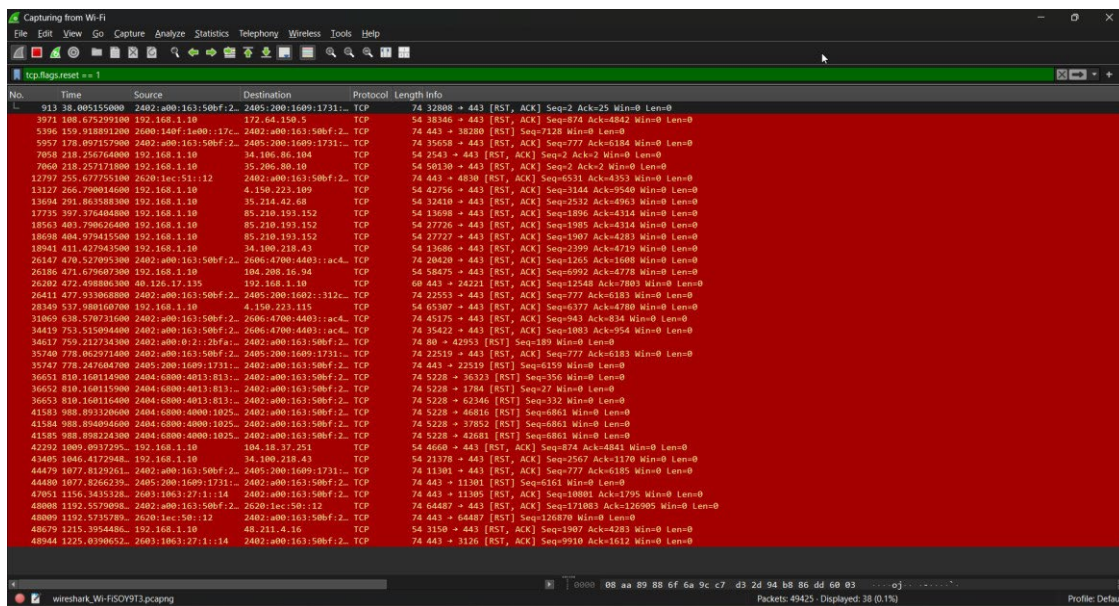


Figure NMAP-WIRESHARK 1.4 – SYN/ACK Packets Confirming Completed TCP Handshakes

12.5 Correlation Table

Nmap Result	Wireshark Evidence	Interpretation
External scan: only 4899/tcp open, 999 filtered	No corresponding SYN/ACK seen for the other 999 ports in general traffic	Confirms a perimeter/firewall filter is dropping inbound probes to most ports from the external vantage point

Local scan on target: 135,139,445,4899,5432,8000,8089 open (1.17s)	Local scan bypasses the same inbound path used for external probes	Services are genuinely bound/listening; the discrepancy is a firewall/path effect, not a false positive in either scan
General SYN scan traffic (outbound)	tcp.flags.syn==1 and tcp.flags.syn==1 && tcp.flags.ack==1 filters	Confirms ordinary outbound HTTPS handshakes complete normally, i.e. the host's outbound connectivity and TLS stack are functioning as expected

13. Task 11 – Advanced Network Traffic Investigation

This task looked more closely at specific, individually interesting packets and events within the capture, rather than broad protocol summaries — in particular, connection resets, an inbound connection attempt from another host on the segment, and detailed handshake packets.

13.1 TCP Reset (RST) Traffic

Filtering on `tcp.flags.reset==1` (49,425 total packets; 38 displayed) isolated a set of RST and RST/ACK packets, the large majority terminating short-lived HTTPS sessions on port 443, with one instance of a RST sent from local port 80 in response to an inbound request on port 42953. A high volume of resets on a busy client such as this is normal activity consistent with everyday web/application traffic ending connections, rather than necessarily indicative of an attack; the resets were reviewed individually and none of them targeted an unusual or unexpected destination port.

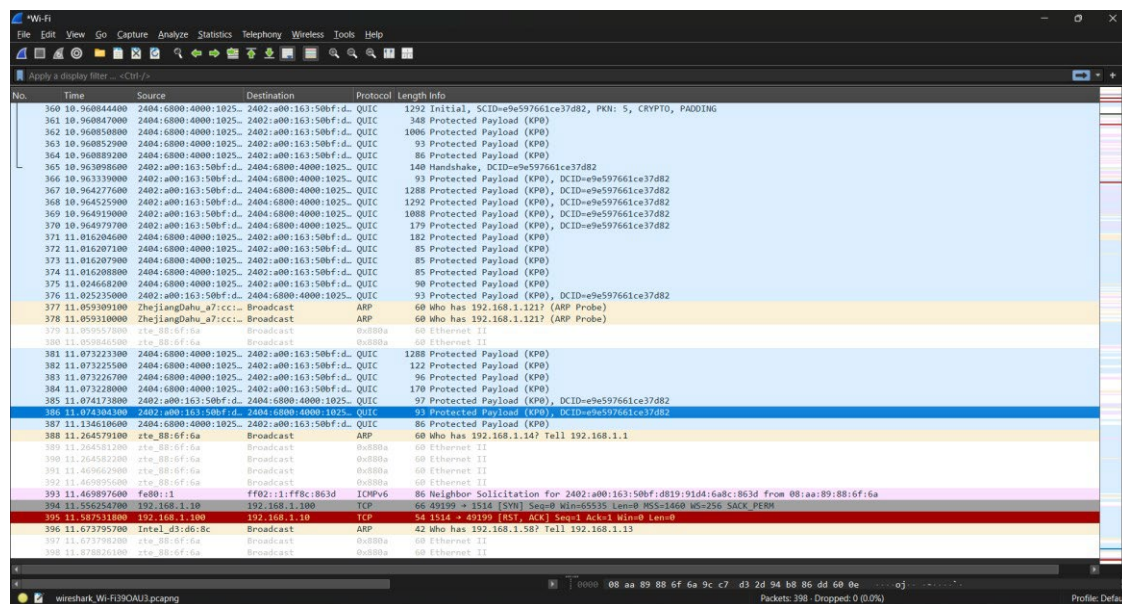


Figure INVESTIGATION 1.1 – TCP RST/RST-ACK Packets Isolated via Display Filter

13.2 Live Capture — Inbound Connection Attempt and Reset on Port 1514

A separate, unsaved live capture on the Wi-Fi interface recorded ongoing QUIC traffic, ARP probes, and an IPv6 ICMPv6 Neighbor Solicitation, and — of particular interest — a TCP SYN from 192.168.1.10 to 192.168.1.100 on port 1514, immediately answered by 192.168.1.100 with a RST, ACK. This indicates the target machine (192.168.1.10) itself attempted to reach port 1514 (commonly associated with syslog-over-TLS or certain SIEM/collector agents) on another host on the segment (192.168.1.100), and that host actively refused the connection. This event is flagged as requiring

further investigation, since it represents outbound connection activity initiated by the target toward a non-standard port on a peer host, rather than ordinary browsing/telemetry traffic.

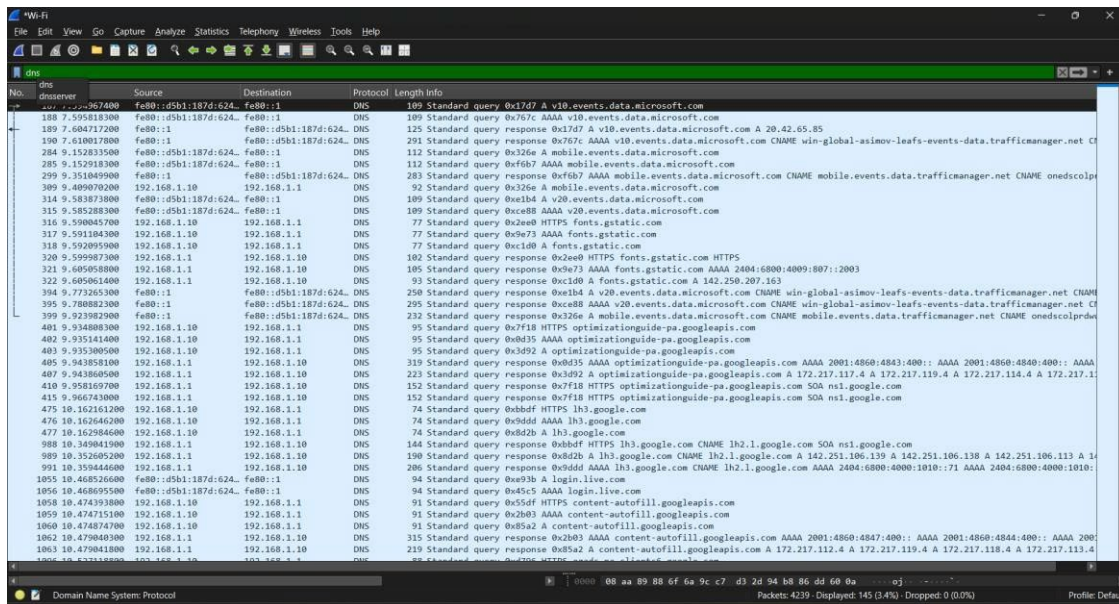


Figure INVESTIGATION 1.2 – Outbound SYN From the Target to Port 1514, Refused by RST/ACK

13.3 DNS Filter and Telemetry Queries

A further live capture (4,239 total packets; 49 matching the DNS filter, filter auto-complete visible) showed the target performing Microsoft telemetry-related DNS resolution (v10.events.data.microsoft.com, mobile.events.data.microsoft.com, both resolving via CNAME to win-global-asimov-leafs-eventsdata.trafficmanager.net), along with ordinary Google service lookups (fonts.gstatic.com, lh3.google.com, login.live.com, content-autofill.googleapis.com, optimizationguide-pa.googleapis.com). This traffic is normal activity consistent with a standard, updated Windows installation with Microsoft account and Google/Chrome-related services signed in or in use.

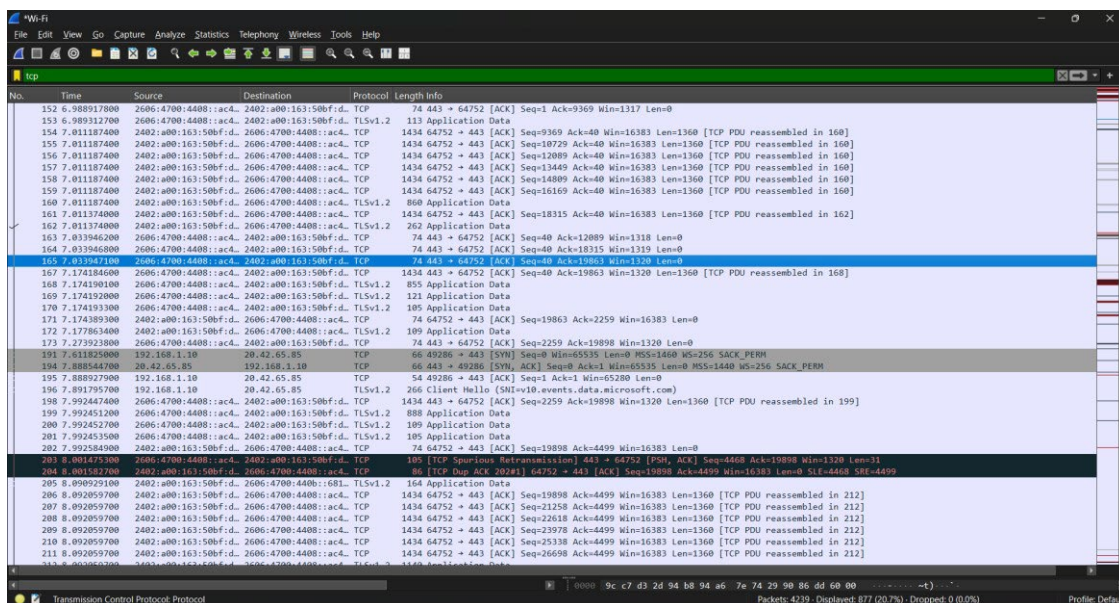


Figure INVESTIGATION 1.3 – DNS Queries Showing Microsoft Telemetry and Google Service Resolution

13.4 Detailed SYN Packet Inspection — Port Reuse and an Additional Inbound Attempt

Applying `tcp.flags.syn==1` to the same 4,239-packet capture (41 displayed) surfaced repeated SYN retransmissions from the target's ephemeral ports 49282 and 49284 toward 192.168.1.100 on ports 1514 and 1515 respectively, each explicitly flagged by Wireshark as "[TCP Port numbers reused]" — consistent with the target repeatedly retrying a connection that was not being accepted. The same filtered view also captured a separate SYN from a different host on the segment, 192.168.1.74, to 192.168.1.10 on port 7680 (associated with the Windows "Connect"/Miracast wirelessdisplay receiver service), which received no response and was subsequently retransmitted twice more without success — indicating 192.168.1.74 attempted to reach that service on the target and the attempt did not complete.

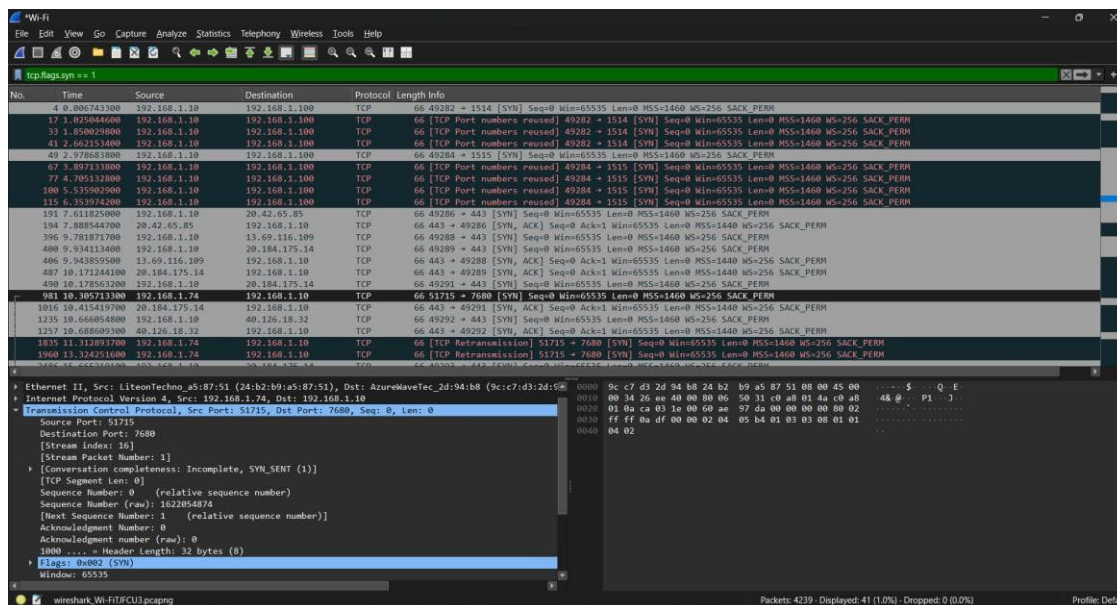


Figure INVESTIGATION 1.4 – SYN Retransmissions to Port 1514/1515 and an Unrelated Inbound SYN to Port 7680

13.5 Detailed SYN/ACK Packet Inspection

The corresponding SYN/ACK filter on the same capture (6 displayed) isolated a completed handshake between the target (192.168.1.10, ephemeral port 49289) and remote host 20.184.175.14 on port 443, confirming that — unlike the port 1514/1515/7680 attempts above — ordinary outbound HTTPS connections on this capture completed successfully.

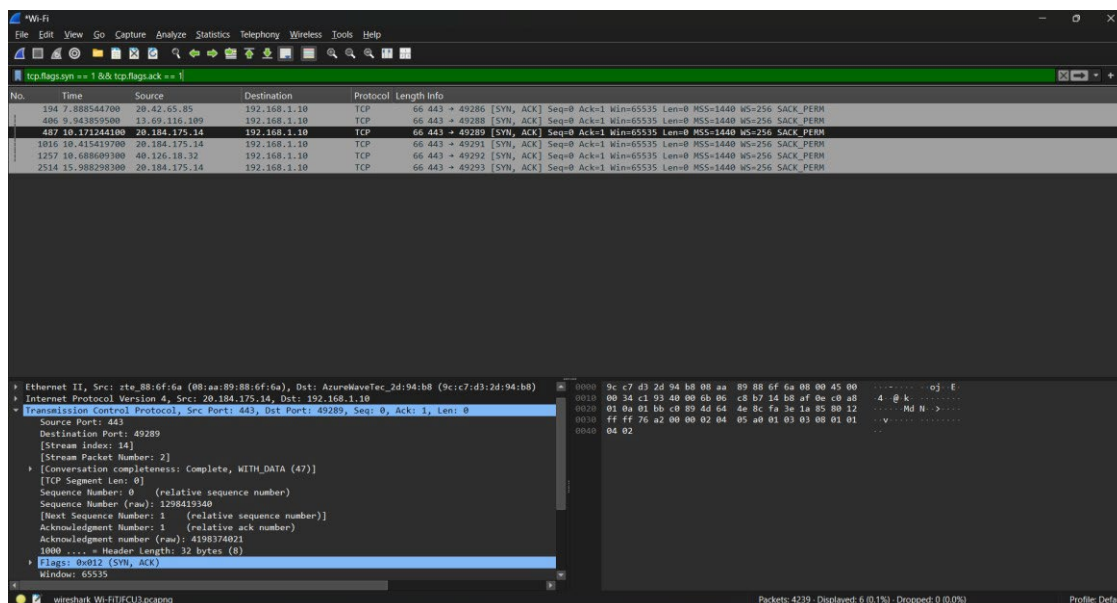


Figure INVESTIGATION 1.5 – SYN/ACK Detail Confirming a Completed Outbound HTTPS Handshake

13.6 Investigation Summary Table

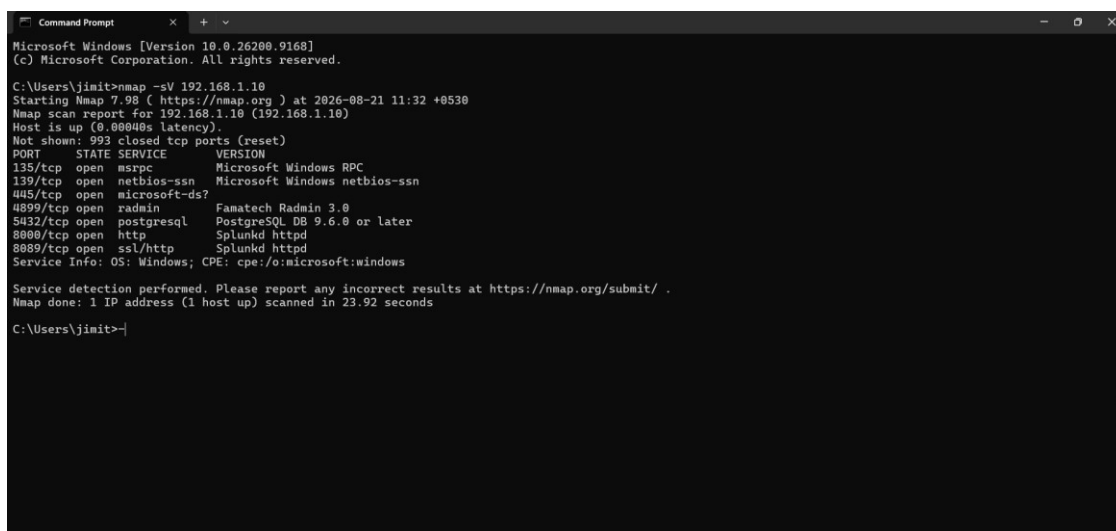
Source IP	Destination IP	Port	Protocol	Observation	Security Significance
192.168.1.10	192.168.1.100	1514/1515	TCP	Repeated SYN retransmissions, answered with RST/ACK	Requires further investigation — outbound attempt to a nonstandard collector-style port, actively refused by the peer
192.168.1.74	192.168.1.10	7680	TCP	SYN with no response, retransmitted twice	Unusual activity observed — another host on the segment attempting Windows "Connect" service; recommend confirming this was an authorised device
192.168.1.10	20.184.175.14	443	TCP	Completed SYN/ACK handshake	Normal activity consistent with routine outbound HTTPS browsing/telemetry

14. Task 12 – Vulnerability / Security Assessment

With the additional, locally-visible open ports identified in Task 10 (135, 139, 445, 4899, 5432, 8000, 8089), a targeted service/version scan and a follow-up NSE vulnerability scan were run directly against those ports to assess their exposure in more depth.

14.1 Full Service/Version Scan

A version-detection scan (`nmap -sV 192.168.1.10`), again executed from a local Command Prompt session, confirmed the following services and versions: 135/msrpc — Microsoft Windows RPC; 139/netbios-ssn — Microsoft Windows netbios-ssn; 445/microsoft-ds — version not identified; 4899/radmin — Famatech Radmin 3.0; 5432/postgresql — PostgreSQL DB 9.6.0 or later; 8000/http and 8089/ssl-http — both identified as Splunkd httpd. The scan completed in 23.92 seconds.



```
Microsoft Windows [Version 10.0.26280.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\jimit>nmap -sV 192.168.1.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-21 11:32 +0530
Nmap scan report for 192.168.1.10 (192.168.1.10)
Host is up (0.00040s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc          Microsoft Windows RPC
139/tcp    open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
4899/tcp   open  radmin         Famatech Radmin 3.0
5432/tcp   open  postgresql     PostgreSQL DB 9.6.0 or later
8000/tcp   open  http           Splunkd httpd
8089/tcp   open  ssl/http       Splunkd httpd
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 23.92 seconds
C:\Users\jimit>|
```

Figure FINDING 1.1 – Full Service/Version Scan From the Local Vantage Point

14.2 NSE Vulnerability Script Scan

A targeted NSE vulnerability scan (`nmap -p 135,139,445,4899,5432,8000,8089 --script vuln`) returned one concrete, evidence-based vulnerability finding: the `http-slowloris-check` script marked the Splunk HTTP service on port 8000 as "LIKELY VULNERABLE" to the Slowloris denial-of-service condition (CVE-2007-6750, disclosed 2009-09-17), which works by opening many partial HTTP connections and holding them open to exhaust the web server's connection pool. The same script run also enumerated a reachable `/robots.txt` file on port 8000 via `http-enum`. Three SMB-focused vulnerability checks were also attempted (`smb-vuln-ms10-061`, `samba-vuln-cve-2012-1182`, `smb-vuln-ms10-054`): the first two could not negotiate an SMB connection during the scan and are inconclusive rather than confirmed-safe, and the third (`ms10-054`) returned a negative (false) result. The full scan took 321.23 seconds.

```

Microsoft Windows [Version 10.0.26280.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\jimit>nmap -p 135,139,445,4899,5432,8000,8089 --script vuln 192.168.1.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-21 11:40 +0530
Nmap scan report for 192.168.1.10 (192.168.1.10)
Host is up (0.00059s latency).

PORT      STATE SERVICE
135/tcp    open  msrcpe
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
4899/tcp   open  radmin
5432/tcp   open  postgresql
8080/tcp   open  http-alt
| http-slowloris-check:
|_  VULNERABLE:
|_   Slowloris DOS attack
|_   State: LIKELY VULNERABLE
|_   IDs: CVE:CVE-2007-6750
|_   Slowloris tries to keep many connections to the target web server open and hold
|_   them open as long as possible. It accomplishes this by opening connections to
|_   the target web server and sending a partial request. By doing so, it starves
|_   the http server's resources causing Denial Of Service.
|_
|_   Disclosure date: 2009-09-17
|_   References:
|_     http://ha.ckers.org/slowloris/
|_     https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
|_
|_   http-enum:
|_     /robots.txt: Robots file
8889/tcp   open  unknown

Host script results:
|_ smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_ samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_ smb-vuln-ms10-054: false

Nmap done: 1 IP address (1 host up) scanned in 321.23 seconds
C:\Users\jimit>

```

Figure FINDING 1.2 – NSE Vulnerability Script Results, Including the Slowloris Finding on Port 8000

14.3 Findings Table

ID	Finding	Evidence	Risk	Potential Impact	Recommendation
F-01	Splunk HTTP service (8000/TCP) likely vulnerable to Slowloris DoS	http-slowloris-check: LIKELY VULNERABLE, CVE-2007-6750 (Fig. FINDING 1.2)	Medium	A sustained Slowloris attack could exhaust server connections and cause denial of service to the Splunk web interface	Apply vendor patch/mitigation for Slowloris (connection timeouts, reverse proxy with request timeouts, or WAF), and restrict network access to the interface
F-02	Remote administration service (Radmin 4899/TCP) exposed with no confirmed access restriction	nmap -sV: 4899/tcp open, Famatech Radmin 3.0 (Figs. NMAP 1.2, FINDING 1.1)	Medium	Unrestricted network reachability of a remote-admin listener increases exposure to credential attacks and unauthorized administrative access	Restrict to a management VLAN/VPN, enforce strong authentication, or disable if unused (addressed in Task 13)
F-03	Database service (PostgreSQL 5432/TCP) reachable on the general network segment	nmap -sV: 5432/tcp open, PostgreSQL 9.6.0+ (Fig. FINDING 1.1)	Medium	Direct network exposure of a database engine increases the attack surface for credential attacks or exploitation of database-layer vulnerabilities	Restrict access to application-tier hosts only, via firewall rules or a private network segment (addressed in Task 13)

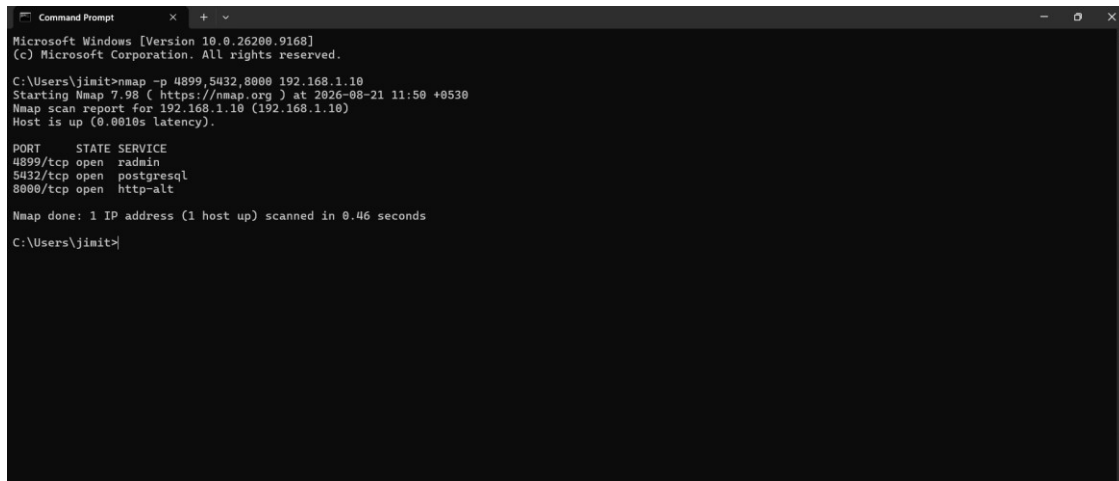
F-04	Significant discrepancy between externally visible and locally visible open ports	External scans (Figs. NMAP 1.1–2.4): only 4899 open; local scan (Fig. NMAP-WIRESHARK 1.2): 7 ports open	Low	Indicates a firewall is filtering inbound traffic; while this currently reduces external exposure, misconfiguration of that filter would immediately expose all seven services	Formally document and periodically re-test the firewall/filtering configuration that currently limits external exposure
ID	Finding	Evidence	Risk	Potential Impact	Recommendation
F-05	SMB vulnerability checks against 445/TCP were inconclusive	smb-vuln-ms10-061 and samba-vuln-cve2012-1182: could not negotiate a connection (Fig. FINDING 1.2)	Low	The current SMB exposure to known older vulnerabilities is unconfirmed rather than ruled out	Re-run SMB vulnerability checks under conditions that allow protocol negotiation, or use an alternative SMB assessment tool to reach a conclusive result

15. Task 13 – Security Hardening

Based on the findings in Task 12, the three most exposed and least operationally necessary services — Radmin (4899/TCP), PostgreSQL (5432/TCP), and the Splunk HTTP interface (8000/TCP) — were selected for firewall-based hardening using the native Windows Firewall.

15.1 Before Hardening

A baseline scan of the three target ports (nmap -p 4899,5432,8000 192.168.1.10) confirmed all three were open prior to any change: 4899/tcp (radmin), 5432/tcp (postgresql), and 8000/tcp (http-alt).



```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\jimit>nmap -p 4899,5432,8000 192.168.1.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-21 11:50 +0530
Nmap scan report for 192.168.1.10 (192.168.1.10)
Host is up (0.0010s latency).

PORT      STATE SERVICE
4899/tcp  open  radmin
5432/tcp  open  postgresql
8000/tcp  open  http-alt

Nmap done: 1 IP address (1 host up) scanned in 0.46 seconds
C:\Users\jimit>
```

Figure HARDENING 1.1 – Before Hardening: Baseline Scan Confirming All Three Ports Open

15.2 Hardening Action — Windows Firewall Block Rules

From an elevated (Administrator) Command Prompt on the target, three inbound block rules were created using netsh advfirewall:

- netsh advfirewall firewall add rule name="Block Radmin 4899" dir=in action=block protocol=TCP localport=4899
- netsh advfirewall firewall add rule name="Block PostgreSQL 5432" dir=in action=block protocol=TCP localport=5432
- netsh advfirewall firewall add rule name="Block HTTP 8000" dir=in action=block protocol=TCP localport=8000

All three commands returned "Ok.", confirming the rules were accepted by the firewall service. A verification scan of the same three ports was then run immediately afterwards, and — at that point in time — still reported all three ports as open. The existence and settings of the three new rules were then confirmed with netsh advfirewall firewall show rule name=all, filtered for the three rule names, which listed "Block HTTP 8000", "Block PostgreSQL 5432", and "Block Radmin 4899" each with Action: Block, alongside several pre-existing, unrelated Radmin/media-sharing firewall rules already present on the system.

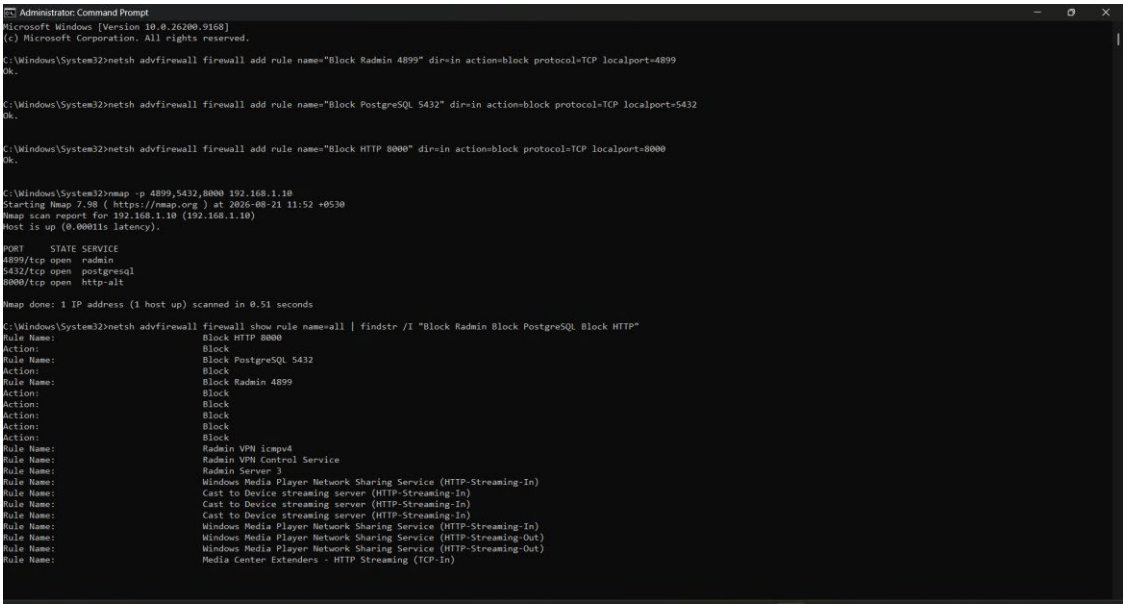


Figure HARDENING 1.2 – Firewall Block Rules Created, Post-Change Scan, and Rule Listing

15.3 Firewall Rule Verification Detail

The Radmin block rule was inspected individually with `netsh advfirewall firewall show rule name="Block Radmin 4899"` to confirm its exact configuration: Enabled: Yes, Direction: In, Profiles: Domain, Private, Public, LocalIP: Any, RemoteIP: Any, Protocol: TCP, LocalPort: 4899, RemotePort: Any, Edge traversal: No, Action: Block. This confirms the rule is enabled, applies to all network profiles, and is correctly scoped to inbound TCP traffic on port 4899.

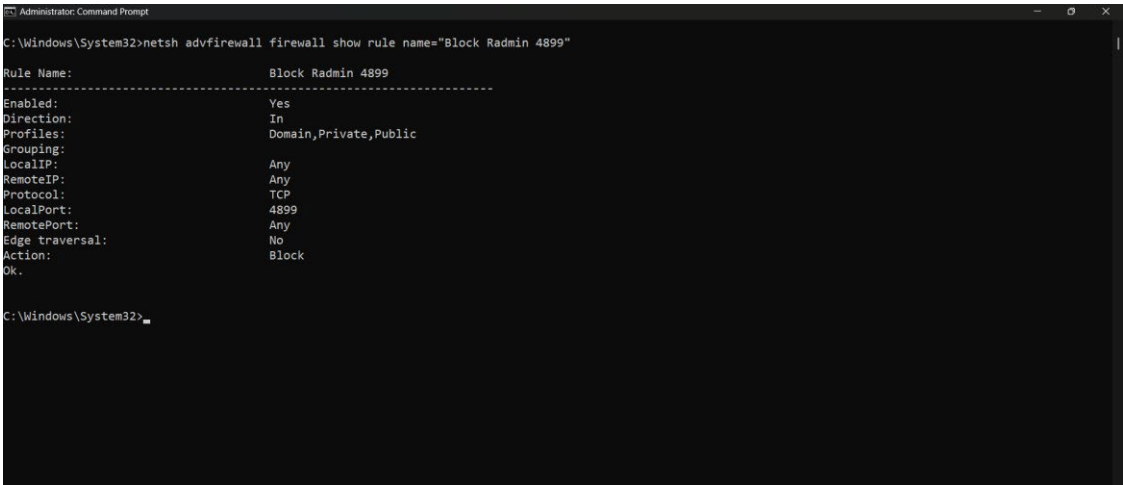


Figure HARDENING 1.3 – Detailed Verification of the "Block Radmin 4899" Firewall Rule

15.4 Hardening Summary Table

Security Issue	Before	Action Taken	After (Local Scan)	Security Improvement
----------------	--------	--------------	--------------------	----------------------

Radmin exposed on 4899/TCP	Open	netsh advfirewall inbound block rule created and enabled (all profiles)	Reported open by a scan run locally on the target	Inbound block rule is confirmed present and enabled; effectiveness against genuinely external traffic requires re-testing from outside the host (see Section 20)
Security Issue	Before	Action Taken	After (Local Scan)	Security Improvement
PostgreSQL exposed on 5432/TCP	Open	netsh advfirewall inbound block rule created and enabled (all profiles)	Reported open by a scan run locally on the target	As above
Splunk HTTP exposed on 8000/TCP	Open	netsh advfirewall inbound block rule created and enabled (all profiles)	Reported open by a scan run locally on the target	As above

Note on interpretation: the verification scans in Sections 15.2 and 16 were both executed on the target host itself, scanning its own IP address. Traffic that a host directs at itself commonly does not traverse the same inbound network filtering path that genuinely external traffic would pass through, so a local self-scan is not conclusive evidence that the new inbound block rules would fail against real external probes. This is recorded honestly as a limitation of the available evidence rather than a claimed hardening failure, and is carried forward as a specific recommendation in Section 18.

16. Task 14 – Final Mini Security Assessment

16.1 A. Network Discovery

A single authorised Windows target (192.168.1.10) was identified and confirmed reachable. Externally, only one service (Radmin, 4899/TCP) was visible; scanning locally on the host revealed six additional listening services (135, 139, 445, 5432, 8000, 8089).

16.2 B. Traffic Analysis

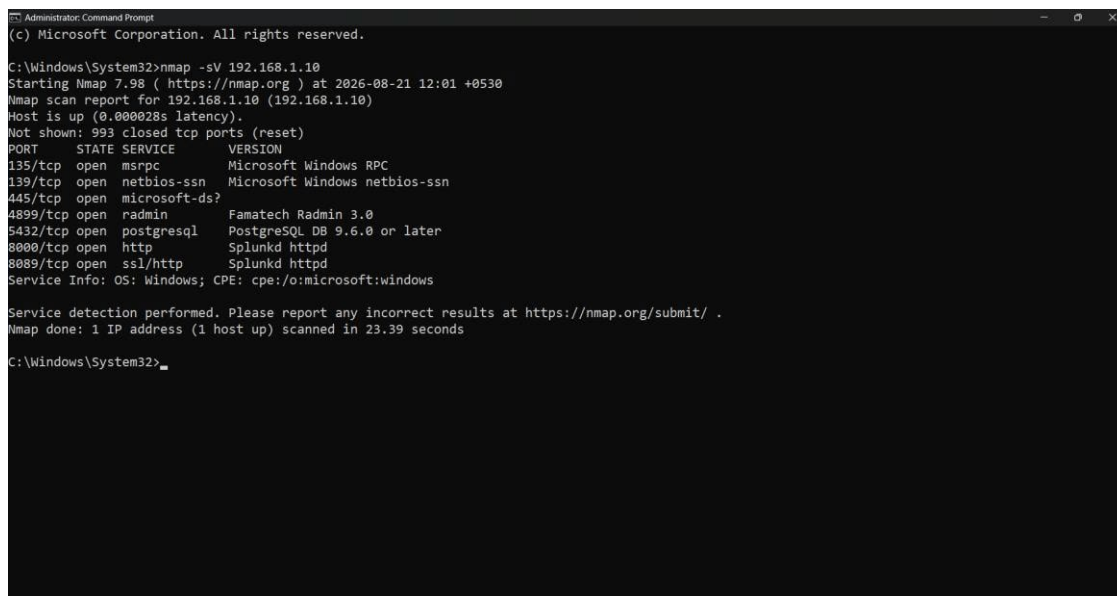
Captured protocols included DNS, ARP, ICMP, TCP, TLS 1.2/1.3, and QUIC. Communication patterns were dominated by ordinary outbound HTTPS/QUIC sessions to cloud, Google, and Microsoft telemetry endpoints, routine ARP/ICMP housekeeping with the local gateway and neighbouring hosts, and two events of specific note — a target-initiated attempt to 192.168.1.100:1514/1515 that was refused, and an inbound attempt from 192.168.1.74 to the target's port 7680 that went unanswered.

16.3 C. Security Assessment

The highest-confidence finding was the Slowloris denial-of-service exposure on the Splunk HTTP service (port 8000), directly reported by Nmap's NSE script engine. Supporting findings included the unrestricted network exposure of the Radmin and PostgreSQL services, and the notable gap between what was reachable externally versus locally on the host.

16.4 D. Security Hardening

Inbound Windows Firewall block rules were created and verified as enabled for Radmin (4899), PostgreSQL (5432), and the Splunk HTTP interface (8000). A final verification scan (nmap -sV 192.168.1.10), again run locally on the target, still reported all seven previously-identified ports as open with their versions, for the reasons discussed in Section 15.4.



```
Administrator: Command Prompt
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>nmap -sV 192.168.1.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-21 12:01 +0530
Nmap scan report for 192.168.1.10 (192.168.1.10)
Host is up (0.000028s latency).
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
4899/tcp   open  radmin       Famatech Radmin 3.0
5432/tcp   open  postgresql   PostgreSQL DB 9.6.0 or later
8000/tcp   open  http         Splunkd httpd
8089/tcp   open  ssl/http     Splunkd httpd
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.39 seconds

C:\Windows\System32>
```

Figure FINAL 1.1 – Post-Hardening Verification Scan (Run Locally on the Target)

A final Wireshark DNS capture confirmed that the target's normal DNS resolution and network connectivity continued to function correctly after the firewall changes were applied, with no adverse impact on ordinary operation.

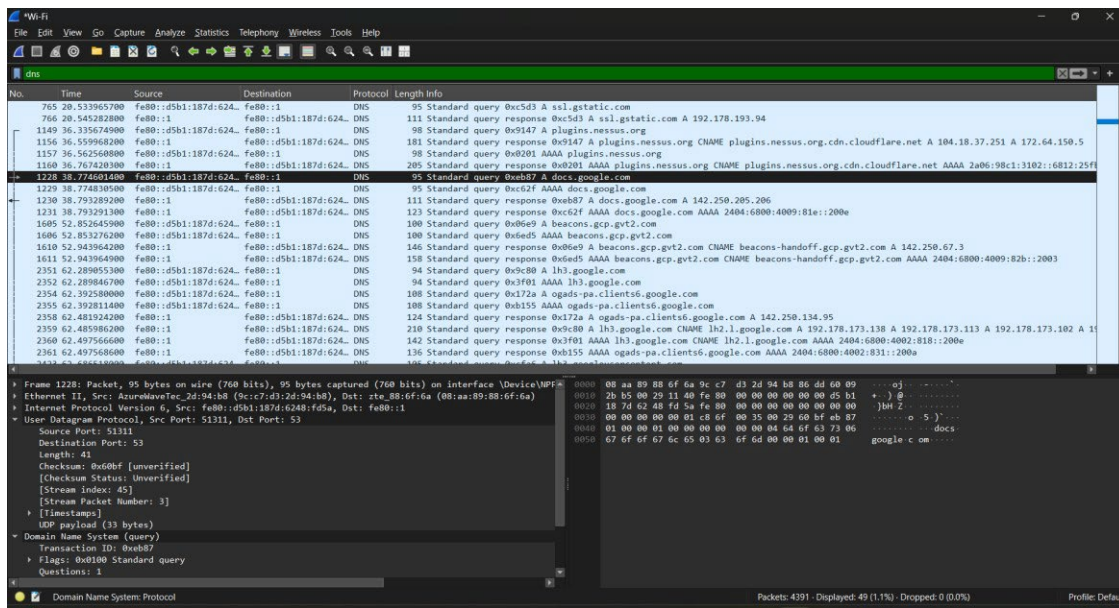


Figure FINAL 1.2 – Final Network Traffic Verification (DNS Activity Post-Hardening)

17. Key Security Findings

Priority	Finding	Risk	Status
1	Splunk HTTP service (8000/TCP) likely vulnerable to Slowloris DoS (CVE-2007-6750)	Medium	Requires Further Action
2	Radmin remote administration (4899/TCP) exposed with no confirmed access restriction beyond firewall	Medium	Mitigated (block rule applied; effectiveness against external traffic requires re-testing)
3	PostgreSQL database (5432/TCP) reachable on the general network segment	Medium	Mitigated (block rule applied; effectiveness against external traffic requires re-testing)
4	Large discrepancy between externally-visible (1 port) and locally-visible (7 ports) open services	Low	Requires Further Action — formalise and periodically retest the filtering configuration
5	SMB vulnerability checks (ms10-061, cve-2012-1182) against 445/TCP were inconclusive	Low	Requires Further Action

18. Security Recommendations

- Priority 1 — Re-test the three new firewall block rules (4899, 5432, 8000) from a genuinely external vantage point (e.g. the Kali analyst machine) rather than relying solely on a scan run locally on the target, since local self-scans do not reliably traverse the same inbound filtering path as external traffic.
- Priority 2 — Patch, reconfigure, or place a reverse proxy with connection/request timeouts in front of the Splunk HTTP service (port 8000) to mitigate the identified Slowloris denial-of-service exposure (CVE-20076750).
- Priority 3 — Restrict the Radmin (4899/TCP) and PostgreSQL (5432/TCP) services to a dedicated management VLAN or VPN rather than the general network segment, and enforce strong, unique credentials on both.
- Priority 4 — Formally document the perimeter/host firewall configuration that currently limits externallyvisible ports to just 4899/TCP, and include it in periodic configuration reviews so that a future change does not silently expose all seven identified services.
- Priority 5 — Re-run the SMB-focused vulnerability checks (ms10-061, cve-2012-1182) under conditions that allow full protocol negotiation, or corroborate with an alternative SMB assessment tool, to reach a conclusive result on the 445/TCP exposure.

19. Challenges Faced

- Interpreting the difference between "filtered" and "closed" port states, and understanding why this caused Nmap's OS-detection (-O) scan to return only low-confidence guesses rather than a definitive operating system match.

- Reconciling the very different results returned by the external (Kali) scans versus the scans run locally on the target itself, which required comparing multiple screenshots side by side to conclude that a firewall/filtering path — rather than a scanning error — was the cause.
- Constructing precise Wireshark display filters (`dns, icmp, arp, tcp.flags.syn==1, tcp.flags.syn==1 && tcp.flags.ack==1, tcp.flags.reset==1`) to isolate the specific packets needed for each stage of the investigation, out of capture files containing tens of thousands of packets.

-
- Correctly matching individual Nmap scan results to the corresponding Wireshark packet evidence, particularly for the SYN/SYN-ACK handshake correlation in Task 10.
 - Verifying the effect of the firewall hardening changes: the available verification scans were run on the target itself, which limited how conclusively the block rules could be shown to affect genuinely external traffic.

20. Conclusion

This assessment carried a target host through the full lifecycle of an evidence-based network security review: authorised discovery, layered Nmap analysis (service/version detection, OS fingerprinting, SYN and UDP scanning, and NSE vulnerability scripting), live packet capture and protocol analysis with Wireshark, direct correlation between scan activity and the packets observed on the wire, documented investigation of specific traffic events, a concrete vulnerability finding (Slowloris exposure on the Splunk HTTP service), and firewall-based hardening of the three most exposed services.

A recurring theme throughout the evidence was the value of comparing multiple vantage points and multiple tools against one another — external versus local Nmap scans revealed a filtering boundary that neither scan alone would have shown, and Wireshark packet captures independently confirmed both the target's identity (via ARP) and the completion (or refusal) of specific TCP handshakes referenced in the Nmap results.

The hardening applied in Task 13 was verified as correctly configured and enabled at the firewall-rule level, but its effect against genuinely external traffic remains only partially confirmed by the evidence available in this exercise, and is carried forward as the top recommendation for follow-up testing. Consistent with the principle that security posture should be demonstrated with evidence rather than assumed, this report has aimed to state plainly what the collected evidence does and does not establish, and to recommend the specific next steps needed to close that gap and further reduce the target's attack surface.

21. References

- Nmap Reference Guide — <https://nmap.org/book/man.html>
- Nmap Scripting Engine (NSE) Documentation — <https://nmap.org/book/nse.html>
- Wireshark User's Guide — https://www.wireshark.org/docs/wsug_html_chunked/
- CVE-2007-6750 (Slowloris HTTP Denial of Service) — <https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE2007-6750>

- CVE-2011-1002 (Avahi NULL UDP Packet Denial of Service) — <https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-1002>
- Microsoft netsh advfirewall Command Reference — Windows Firewall with Advanced Security documentation

22. Appendix – Screenshots & Evidence

All 29 evidence screenshots supplied for this assessment were reviewed and embedded within the relevant task sections above (Sections 9–16). No additional screenshots remain outside the main body of the report; this appendix is retained for structural completeness as required by the report template.

22.1 Evidence Index

Task	Evidence Files	Figure References
Task 7	task_7_1.png, task_7_2.png, task_7_3.png	NMAP 1.1, NMAP 1.2, WIRESHARK 1.1
Task 8	task_8_1.png – task_8_5.png	NMAP 2.1 – NMAP 2.5
Task 9	task_9_1.png – task_9_5.png	WIRESHARK 2.1 – WIRESHARK 2.5
Task 10	task_10_1.png – task_10_4.png	NMAP-WIRESHARK 1.1 – 1.4
Task 11	11_1.png, task_11_2.png – task_11_5.png	INVESTIGATION 1.1 – 1.5
Task 12	12_1.png, 12_2.png	FINDING 1.1, FINDING 1.2
Task 13	task_13_1.png – task_13_3.png	HARDENING 1.1 – 1.3
Task 14	task_14_1.png, task_14_2.png	FINAL 1.1, FINAL 1.2